

Predicting Jeopardy! Winners from Contestant–Board Semantics

Group: SLMB: Statistics, Learning, Models, and Beyond

Leader: Shriya Ramaka

Members: Lyle Mora, Maggie Parkhurst, and Bhargav Garre Venkata

Course: DS 4002, Section 002 | Date: September 11, 2026

PROJECT GOAL

For each historical Jeopardy! game, we will use the three contestants' occupation-based semantic profiles and the game's clue board to predict the winner, evaluating success by whether the model achieves lower log loss on held-out games than a matched non-semantic model and uniform one-in-three probabilities.

RESEARCH QUESTION

Given the three contestants and the clues on an actual historical Jeopardy! board, does semantic alignment between each contestant's occupational profile and the board improve prediction of the recorded winner?

MOTIVATION

Jeopardy! rewards broad knowledge, rapid recall, buzzer skill, and strategy, while contestants are introduced partly through their occupations. Occupations may reflect specialized vocabulary and subject knowledge developed through education and work. Because each board contains a different mixture of history, science, literature, popular culture, and other topics, the same occupational background may be more relevant to some games than to others. Text and subject matter have also helped explain trivia-question difficulty [1]. This motivates a retrospective, board-conditioned question: without using gameplay results, can the semantic match between each contestant's occupational profile and a revealed board help identify the eventual winner? The outcome is the recorded winner rather than a manually assigned label, which makes performance directly testable against real historical results. The analysis will connect clue and scoring records [2] with contestant occupations and winners from Cluebase [3].

An initial audit found 6,308 games with exactly three contestants, one winner, and complete contestant names and notes; 6,304 also contain clue-board text and form the current analyzable sample. Four games without board text will be excluded, and unusually incomplete boards will be examined in a sensitivity analysis. Coryat scores, which measure Jeopardy! clue performance by adding the values of correct responses and subtracting the values of incorrect non-Daily Double responses, reducing the influence of wagering [4], may be summarized descriptively but will not be used to predict the winner. The project tests whether semantic information contributes anything beyond simpler comparisons. A positive result would mean that occupation-to-board alignment improves out-of-sample winner probabilities, not that occupation alone determines success. A weak or null result would also be informative because winning depends on factors that text cannot capture, including response speed, clue selection, Daily Double placement, and wagering. As a separate exploratory application, UVA major and course descriptions [5], [6] and the CIP-SOC Crosswalk, a method of connecting institutional programs to similar occupational fields [7], may be converted into major profiles and compared with boards. Those rankings will represent textual alignment only and will not claim that students in a particular major would win or that education causes Jeopardy! success.

MODELING APPROACH

Occupation phrases will be normalized and linked to SOC descriptions to create one semantic profile per contestant. Sentence-BERT embeddings [8] and cosine similarity will compare each profile with the clue and category text on the same historical board. Average, maximum, category-level, and high-value-clue similarities, plus each contestant's advantage over the other two players, will form the preregistered semantic features. A regularized game-level multinomial logistic model will assign three win probabilities that sum to one. It will be evaluated on held-out historical games using folds grouped by game. Primary evaluation will use log loss; top-one game accuracy, ROC-AUC, and game-level bootstrap intervals will support interpretation. We will compare the semantic model with uniform one-in-three probabilities and a matched model using only permissible non-semantic, pre-outcome characteristics. Winner labels, scores, Coryat, clue responses, and wagers will never be predictors. Accuracy will equal the percentage of held-out boards for which the highest-probability contestant is the recorded winner.

REFERENCES

- [1] E. Boettcher, "Predicting the difficulty of trivia questions using text features," master's paper, School of Information and Library Science, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA, 2016, doi: 10.17615/rmtt-0a51.
- [2] J. Wolle, "Jeopardy clue dataset," GitHub, 2026. [Online]. Available: https://github.com/jwolle1/jeopardy_clue_dataset. [Accessed: Sep. 11, 2026].
- [3] L. Lavin, "What is Cluebase?," Cluebase Documentation. [Online]. Available: <https://cluebase.readthedocs.io/en/latest/>. [Accessed: Sep. 11, 2026].
- [4] JCompete, "Coryat score." [Online]. Available: <https://jcompete.com/coryat-score>. [Accessed: Sep. 11, 2026].
- [5] University of Virginia, "Majors and minors." [Online]. Available: <https://www.virginia.edu/majors-minors/>. [Accessed: Sep. 11, 2026].
- [6] University of Virginia, "Undergraduate record." [Online]. Available: <https://records.ureg.virginia.edu/>. [Accessed: Sep. 11, 2026].
- [7] National Center for Education Statistics, "CIP-SOC crosswalk," 2020. [Online]. Available: <https://nces.ed.gov/ipeds/cipcode/post3.aspx?y=56>. [Accessed: Sep. 11, 2026].
- [8] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in Proc. 2019 Conf. Empirical Methods in Natural Language Processing and 9th Int. Joint Conf. Natural Language Processing (EMNLP-IJCNLP), Hong Kong, China, 2019, pp. 3982–3992, doi: 10.18653/v1/D19-1410.